

Method categories and comparator framework

a Transfer assumptions

1 **Foundation models**
Reusable cell-state representation
scGPT, scFoundation, scBERT

2 **Latent / compositional models**
Latent perturbation effect
scGen, CPA

3 **Prior-graph models**
Regulatory wiring
GEARS, AttentionPert

4 **Optimal-transport models**
Distributional intervention
CellIOT, scPRAM

5 **Hybrid models**
Representation + transition operator
STATE, PertAdapt

6 **Chemistry-aware models**
Compound-structure signal
chemCPA, FP-ridge

b Benchmark roles

1 **Conditioned predictors**
General predictive models evaluated on held-out tasks

2 **Task-specific deterministic predictors**
FP-ridge, linear-shift-KOemb

3 **Distributional comparator**
CINEMA-OT
reference only; never counted as a conditioned win

4 **Baselines**

Universal floor	Context baselines
cell-mean shift linear-PCA shift	control-as-prediction donor shift training-mean shift

c Decision rule

Conditioned prediction

Compare against

cell-mean shift

AND

linear-PCA shift

Counted as working
only if it beats both floor members

Context baselines and task-specific comparators are reported for interpretation

CINEMA-OT
distributional reference only